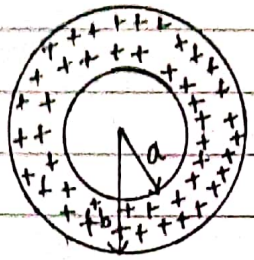
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7C - TNE

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1. Figur 23-53 shows a spherical shell with uniform volume charge density $\rho = 1,84\ nC/m^3$, inner radius $a = 10,0cm$, and outer radius $b = 2.00a$. What is the magnitude of the electric field at radial distance (a) $r = 0$; (b) $r = a/2.00$; (c) $r = a$; (d) $r = 1,50a$; (e) $r = b$, and (f) $r = 3.00b$?



Jawab:

- (a) $r = 0$

$\Rightarrow E = 0$ //

- (b) $r = 2a$

$\Rightarrow E = 0$ //

- (c) $r = a$

$\Rightarrow E = 0$ //

- (d) $r = 1,5a$

$$E = \frac{\rho}{3\varepsilon_0}\frac{r^3 - a^3}{a^2} = \frac{\rho}{3\varepsilon_0}\frac{(1,5a)^3 - a^3}{a^2}$$

$$= \frac{\rho}{3\varepsilon_0}\left(\frac{2,375}{2,25}\right)$$

$$= \frac{(1,84 \times 10^{-9})(0,1)}{3(8,85 \times 10^{-11})}\left(\frac{2,375}{2,25}\right)$$

$$= 7,32\ N/C$$ //

- (e) $r = b$

$$E = \frac{\rho}{3\varepsilon_0}\frac{b^3 - a^3}{b^2} = \frac{\rho}{3\varepsilon_0}\frac{(2a)^3 - a^3}{(2a)^2}$$

$$= \frac{\rho}{3\varepsilon_0}\frac{8a^3 - a^3}{4a^3} \rightarrow \frac{\rho}{3\varepsilon_0}\left(\frac{7}{4}\right)$$

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$= \frac{(1,84 \times 10^{-9})(0,1)}{3(8,85 \times 10^{-12})} \left(\frac{7}{9}\right)$

$= 12,1 \text{ N/C}$ //

- (f) $r = 3b = 6a$

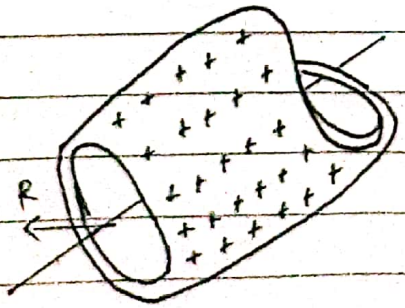
$E = \frac{\rho}{3\varepsilon_0} \frac{b^3 - a^3}{(3b)^2} \Rightarrow \frac{\rho}{3\varepsilon_0} \frac{(2a)^3 - a^3}{(6a)^2}$

$\Rightarrow \frac{\rho}{3\varepsilon_0} \left(\frac{7}{36}\right)$

$\Rightarrow \frac{(1,89 \times 10^{-9})(0,1)}{3(8,85 \times 10^{-12})} \left(\frac{7}{36}\right)$

$\Rightarrow 1,35 \text{ N/C}$ //

2. Figure 23-36 shows a section of a long, thin-walled metal tube of radius $R = 3$ cm, with a charge per unit length of $\lambda = 2 \times 10^{-8}$ C/m. What is the magnitude E of the electric field at radial distance (a) $r = R/2$ and (b) $r = 2R$? (c) Graph E versus r for the range $r = 0$ to $2R$.



- (a) $r = R/2$

$\Rightarrow$ karena $r < R$, $q = 0$, maka $E = 0$

- (b) $r = 2R$

$\Rightarrow$ karena $r > R$, $q = \lambda$ $\lambda = 2 \times 10^{-8}$ C/m $R = 3\text{cm} = 0,03\text{m}$

$r = 0,06$ m

$\Rightarrow E = \frac{\lambda}{2\pi\varepsilon_0}$

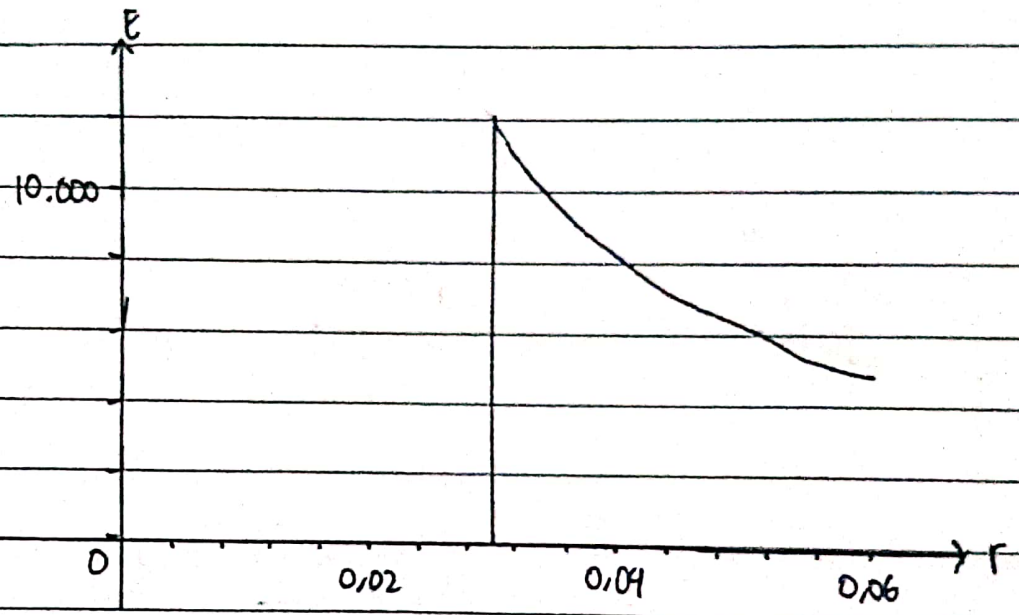
$= \frac{2 \times 10^{-8}}{2\pi(0,06)(8,85 \times 10^{-12})} = 5,99 \times 10^3 \text{ N/C}$ //

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(c) Grafik E vs r



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